



Disorders of Hair

Objectives

- Understand hair growth cycle to be able to evaluate disorders of hair growth
- Review specific types of alopecia
- Identify other causes of hair loss

Hair Growth Basics



- Scalp hair follicles cycle continuously throughout one's life.
 - At any given time, approximately 90% of scalp hairs are in the **growing (anagen) phase** and 10% are in the **resting (telogen) phase**.
 - The duration of anagen determines the length of the hair and lasts up to 7 years or longer to achieve hair lengths of a meter or more.
 - The telogen phase lasts approximately 3 months, after which the resting hairs are shed and new anagen hairs have grow in.
 - Scalp hair grows approximately 0.35 mm/day, or 1 cm/month (1 in. every 2 to 3 months).

Hair Growth Basics

- The scalp typically has about 90,000 to 140,000 hairs (blonds are at the highest end of the range and redheads are at the lowest).
- An average loss of **100 hairs per day is normal**, with more hair lost on days when hair is washed.
- When assessing hair loss, it is **important to know the frequency of washing and when the hair was last washed**.
- Given that dead hairs are removed primarily during washing and brushing, infrequent washing leads to increased incident loss throughout the day.
- If hair has not been washed or brushed for several days, more hair will be removed by gentle hair pull during the physical examination.

Hair Growth Basics



- Examination of the patient should include a routine check for broken hairs and the performance of hair-pull tests on the top, sides, and back of the scalp.
 - The hair-pull test is performed by grasping about 50 to 60 hairs between the index finger and thumb and pulling steadily.
 - Extraction of ≥ 5 telogen hairs in 1 pull or >15 telogen hairs in 3 pulls in someone who has washed his or her hair (best 2-3 days after shampooing)= suggestive of telogen effluvium
- The removed hairs should be telogen hairs, which are recognized by their naked white club-shaped proximal end.
- Anagen hairs are normally difficult to detach and have ragged proximal ends with clear sheaths surrounding their proximal ends.
- Hairs may also break off with pulling.
 - This fragility is usually associated with overprocessing of the hair.

Types of Hair

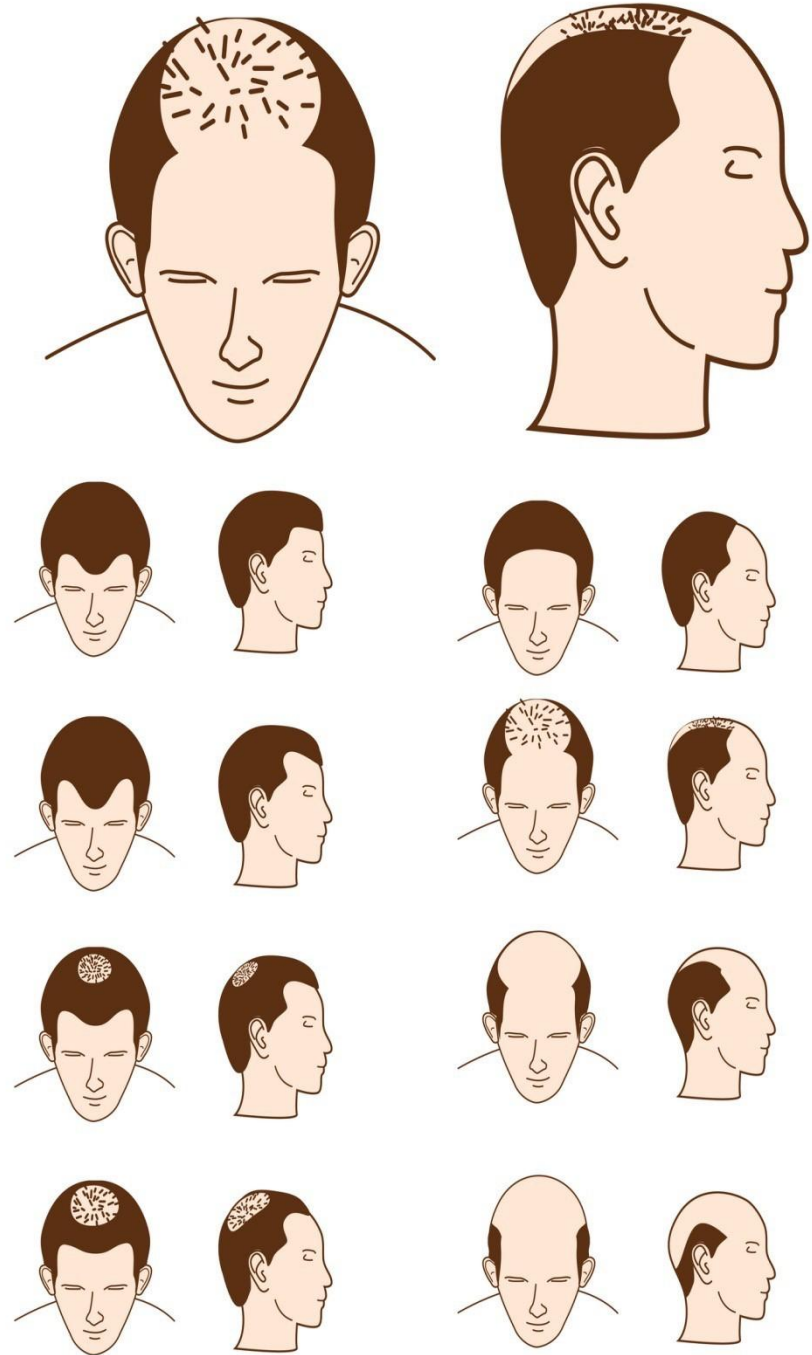
- **Terminal hairs-** Thick, pigmented on the top of the head and in the beard, axillary, and pubic areas are influenced by androgens. Androgens are important in regulating hair growth. At puberty, androgens increase the size of follicles in the beard, chest, and limbs and decrease the size of follicles in the bitemporal region, which reshapes the hairline in men and many women.
- Hair on the rest of the body is independent of androgens.
- **Lanugo** fine hairs found on the fetus and infant
- **Vellus hair** (similar to Lanugo in adults) is short, fine, relatively non-pigmented and covers much of the body.

Pattern Alopecia

Androgenic Alopecia

Pattern Alopecia

- most common type of non-scarring hair loss.
- results from a genetically determined end-organ sensitivity to androgens in follicles within a certain pattern on the scalp.
- It is often referred to as common baldness, male-pattern alopecia, and female-pattern hair loss.

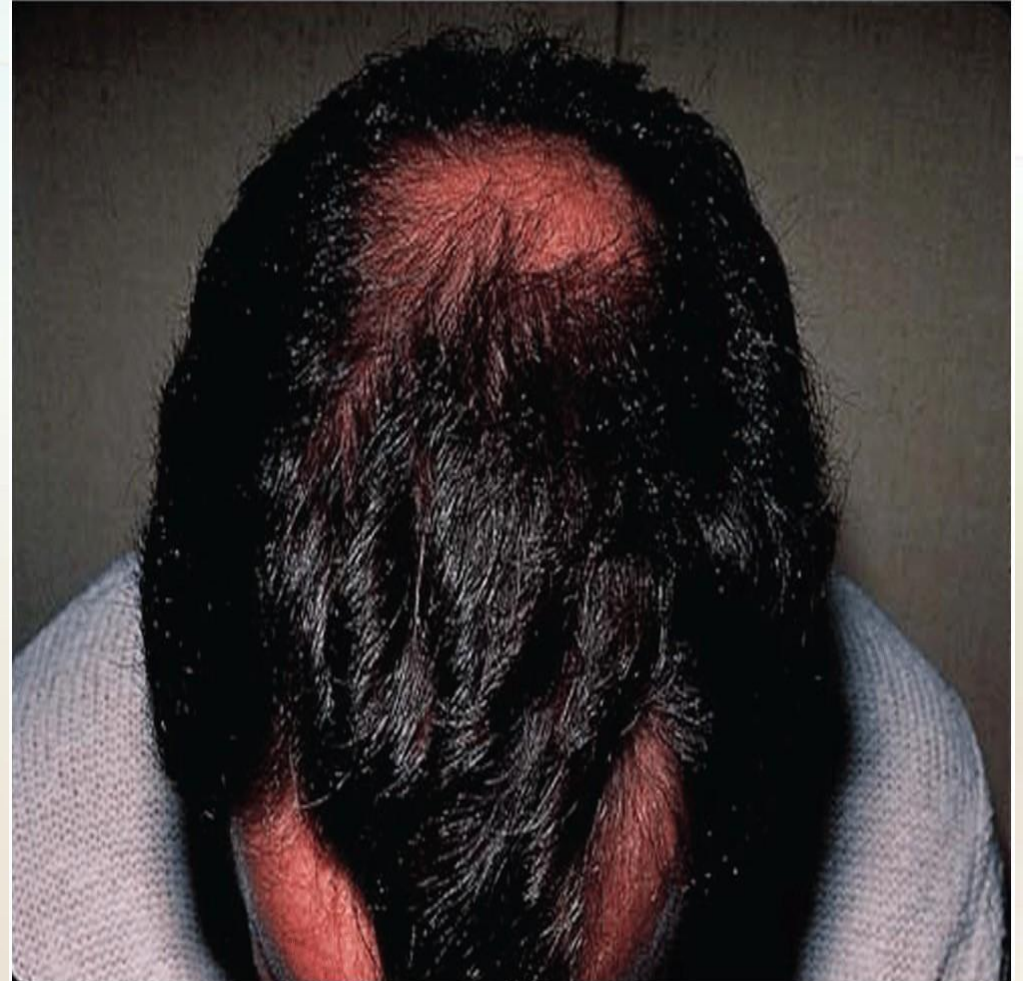


Epidemiology/Pathogenesis

- Affects at least 50% of men by 50 years of age and 40 to 50% of women by 60 years of age.
- Male-pattern alopecia can start between 15 and 25 y/o
- Female-pattern alopecia can start at any point after menarche.
 - The appearance of androgenetic alopecia in women after menopause is common.
- Miniaturization of hairs and decreased hair density account for the appearance of hair loss.
- Androgenetic alopecia largely spares the back and sides of the scalp.

Diagnosis

- The diagnosis of androgenetic alopecia is usually obvious from the **clinical pattern of hair loss**
- Male-pattern alopecia typically manifests with bitemporal recession and vertex balding, which can progress to complete balding of the crown.



Diagnosis

- Female-pattern hair loss - **intact frontal hairline and diffuse thinning over the top of the scalp with widening of the midline part**
- progresses to mild, moderate, or severe thinning but not to complete baldness.
- In women, a male pattern of alopecia occurring with menstrual irregularities, acne, hirsutism, and a deep voice is significant.
 - **Virilism indicates significant hyperandrogenism, the cause of which must be identified and treated**
- Scalp biopsies are rarely necessary to diagnose androgenetic alopecia in men. **In women**, biopsies cut horizontally are useful, especially for diagnosing early androgenetic alopecia.



Treatment - Topical

- **Minoxidil** (now OTC) is approved by the FDA for men over 18 years old but is also effective in adolescents.
 - acts by prolonging anagen and pushing resting hair follicles into the growing phase.
 - produces visible hair growth in approximately one third of male and female patients, fine-hair growth in approximately one third, and no growth in approximately one third.
 - more effective as a preventive agent, retarding hair loss in approximately 80% of patients.
- Side effects include scalp irritation and increased facial hair in women

Treatment - Systemic

- Oral finasteride (propecia) blocks conversion of testosterone to dihydrotestosterone (DHT).
- Hair-weight studies have shown that finasteride increases hair length and diameter, producing better coverage from existing hairs
- Side effects in men are relatively rare but include a lack of libido (< 5%), impotency (< 2%), and a reduction in semen volume (» 10%).
 - These effects are reversed when the drug is stopped and often disappear as the drug is continued.
- may be used in combination with topical minoxidil as they may have a synergistic effect.

Treatment – Therapy for Women

- Topical minoxidil is currently the best available treatment for androgenetic alopecia in women.
- Oral contraceptives can reduce hair loss and occasionally lead to slight hair regrowth, especially in women with irregular menses.
- Oral spironolactone in dosages of 75 to 200 mg/day may be valuable in women with high-normal or elevated levels of DHEAS.
- Iron supplementation should be considered in patients with low serum ferritin (e.g., < 40 ng/mL) or iron deficiency anemia.

Treatment - Surgery

- Hair transplantation
 - involves surgically moving hair follicles from one area of scalp to another
 - To achieve a clinically acceptable outcome donor scalp must have a higher hair density than the recipient area.
 - In general, patients with visible thinned areas and high-density donor areas are good candidates.
 - The procedure can help both men and women, but men are often better candidates.

Table 1 Causes of Diffuse Alopecia¹⁸

Telogen effluvium (acute and chronic)

Anagen arrest

Reactions to drugs and other chemicals

Thyroid disorder

Iron deficiency and other nutritional deficiencies

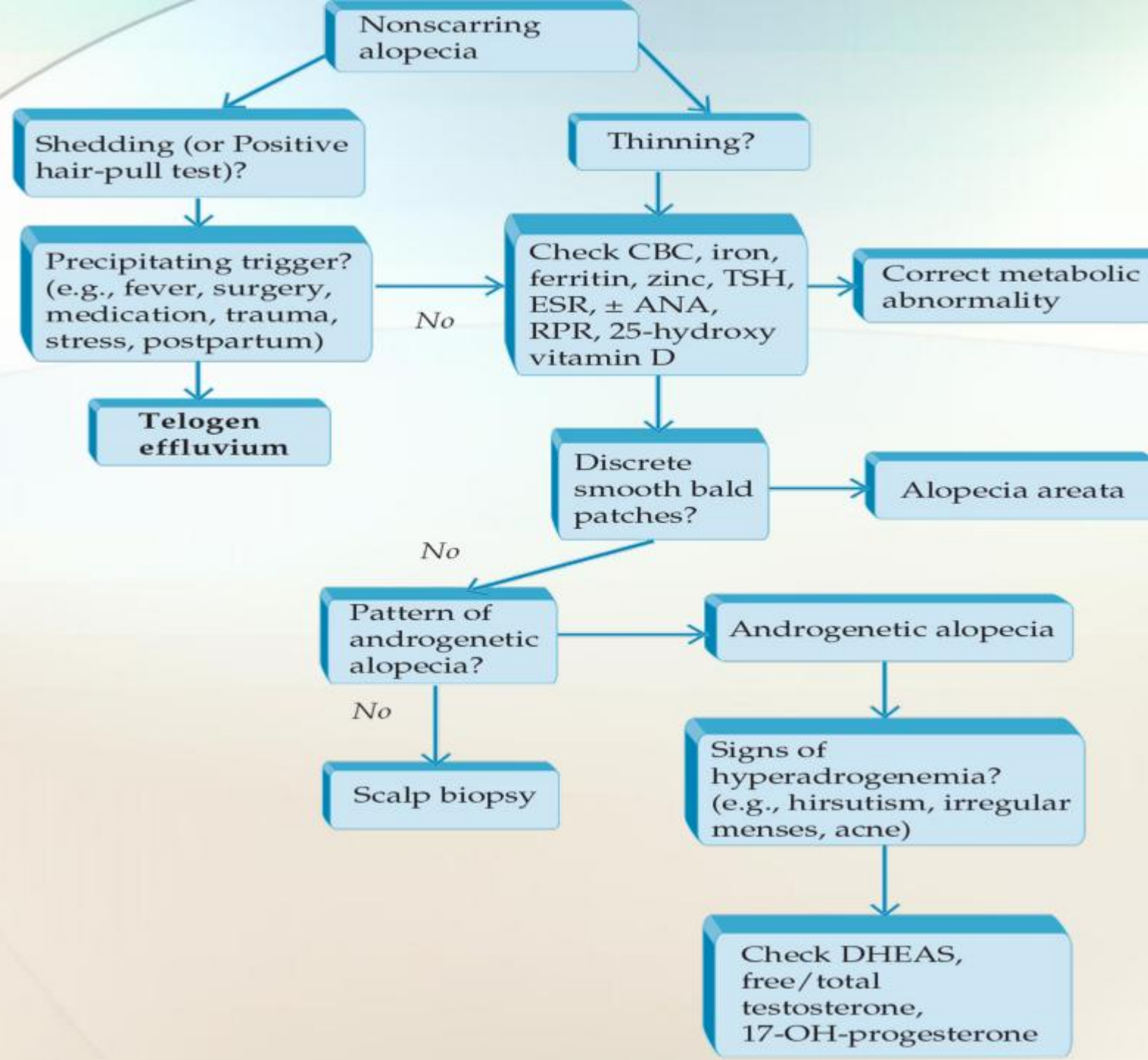
Malabsorption

Renal failure

Hepatic failure

Systemic disease

Miscellaneous causes (e.g., diffuse alopecia areata, congenital hypotrichosis) and idiopathic causes



Diffuse Hair Shedding

Telogen effluvium

Anagen arrest

Drug reactions

Systemic and nonsystemic conditions

Telogen Effluvium

- **Most common form of diffuse alopecia**
- Patients present with generalized shedding of telogen hairs caused by synchronous entry of a large number of follicles into telogen
- **Patients need reassurance - apparent loss of hair is actually a sign of regrowth.**
- Various triggers can precipitate an episode of acute telogen effluvium
 - Many medications may result in diffuse hair shedding
 - The acute form (due to a precipitating trigger) **usually ends within 3 to 6 months.**



Telogen Effluvium

- **Postpartum effluvium** is caused by prolonged anagen duration during pregnancy, with synchronous entry of many follicles into telogen postpartum.
- **Usually stops within 2 to 6 months after delivery, although regrowth takes much longer.**
- Telogen effluvium may unmask an underlying predisposition for androgenetic alopecia
 - as the hair regrows, it may miniaturize in genetically predisposed hair follicles.
- Telogen effluvium **is often seen normally in neonates**

Telogen Effluvium - Diagnosis

- **Clinical**
- may have a history of an initiating event 3 months before the onset of shedding
- During acute telogen effluvium, pull tests are positive all over the scalp, yielding 2-10 club hairs.
- A scalp biopsy may be necessary to distinguish from androgenetic alopecia or diffuse alopecia areata.

Telogen Effluvium - Treatment

- **Reassurance**; no treatment is needed as hair usually regrows within a short time.
- In long standing (chronic) telogen effluvium, topical minoxidil may be indicated.
- Chronic telogen effluvium may improve with correction of underlying nutritional deficiency (i.e., iron, zinc or biotin supplementation).

Table 2 Causes of Telogen Effluvium

Medications [*see Table 3*]

Nutrition aberration (protein-calorie restriction or vitamin deficiency/supplementation)

Physiologic stress (e.g., high fever, surgery)

Severe emotional stress

Postpartum

Metabolic alterations (hypothyroidism)

Table 3 Categories of Drugs that Can Cause Alopecia¹⁸

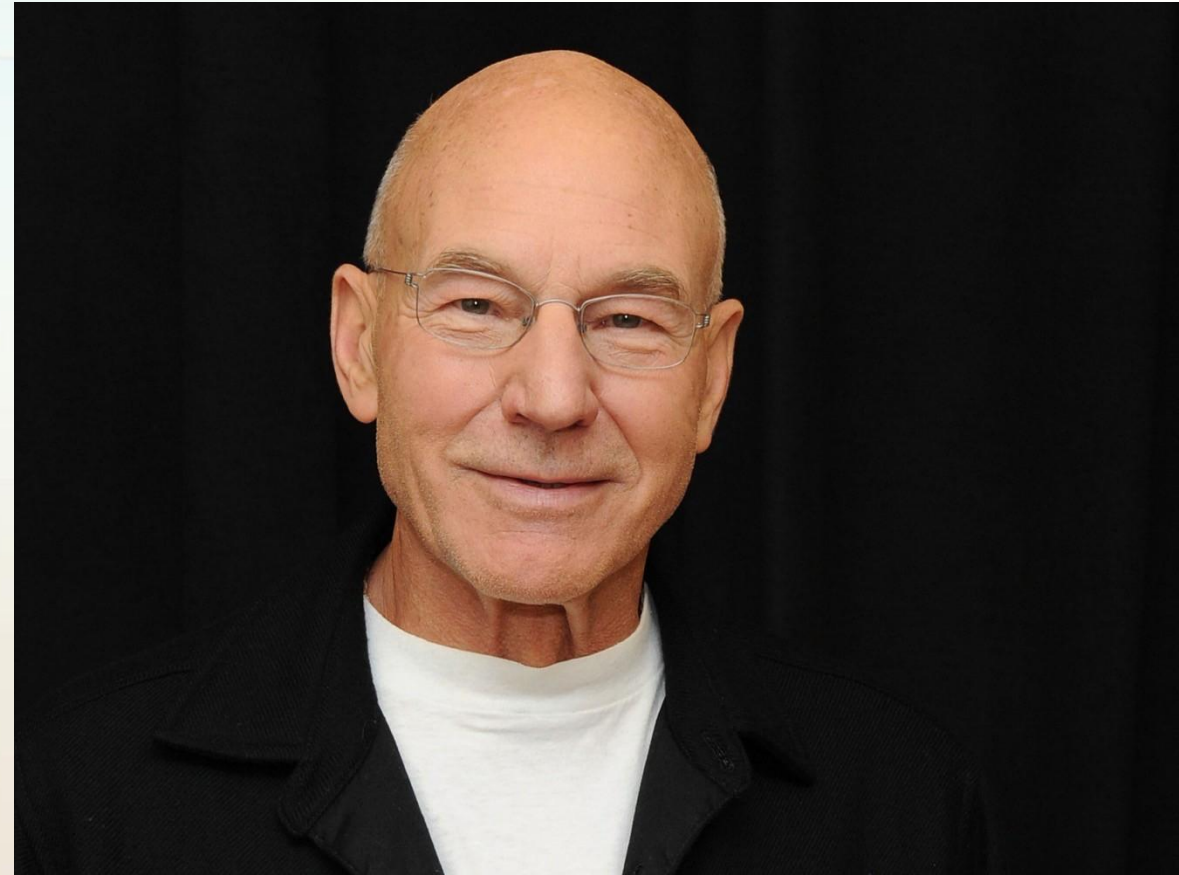
<i>Category</i>	<i>Selected Agents</i>
Alpha blockers	Doxazosin, prazosin, terazosin
Angiotensin-converting enzyme inhibitors	Captopril, enalapril
Anticoagulant drugs	Dicumarol, heparin, warfarin
Anticonvulsant drugs	Ethotoin, mephenytoin, paramethadione, phenytoin, trimethadione, valproate sodium, lithium
Selective serotonin reuptake inhibitors	Fluoxetine, sertraline
Antithyroid drugs	Carbimazole, methylthiouracil, methimazole, propylthiouracil
Beta blockers	Acebutolol, atenolol, labetalol, metoprolol, nadolol, pindolol, propranolol, timolol
Biologic agents	Etanercept, adalimumab, infliximab
Calcium channel blockers	Diltiazem, verapamil
Cholesterol reducers	Clofibrate, lovastatin
H ₂ receptor blockers	Cimetidine, famotidine, ranitidine
Nonsteroidal antiinflammatory drugs	Fenoprofen, ibuprofen, indomethacin, ketoprofen, meclomen, naproxen, piroxicam, sulindac
Retinoids and retinol	Acitretin, etretinate, isotretinoin, vitamin A overdose
Tricyclic antidepressants	Amitriptyline, amoxapine, desipramine, doxepin, imipramine, nortriptyline, protriptyline, trimipramine

Anagen Arrest (Anagen Effluvium)

- Represents a diffuse loss of anagen hairs from growing follicles.
- Inhibition of cell division in the hair bulb matrix leads to a progressive narrowing of the hair shaft and sometimes failure of hair formation.
- **As the growing hair narrows near the skin surface, it may break off.**
 - The resultant shedding can occur within a few weeks, unlike in telogen effluvium, in which shedding takes 3 months to occur.
- **Causes - reactions to cytostatic drugs and other toxic agents, radiation therapy, endocrine diseases, alopecia areata, cicatricial alopecia, trauma and pressure, and severe protein-calorie malnutrition.**
 - Because 90% of scalp hairs are in anagen at any given time, this condition causes obvious and severe baldness.

Alopecia Areata

- **Autoimmune** disease with genetic predisposition
- Typically characterized by **patchy hair loss**
 - Can vary from a single patch on the scalp or elsewhere to total body baldness (alopecia universalis).
- **Favorable prognosis** with many patients regrowing hair spontaneously
- In cases of extensive alopecia areata, alopecia totalis, and alopecia universalis, however, hair loss may be permanent.



Alopecia Areata - Diagnosis

- Characterized by **smooth patches of hair loss**, sometimes surrounded by so-called exclamation-point hairs
 - **not always seen but are diagnostic when present.**
 - They delineate the active spreading margin of alopecia areata.
- Generally affect the scalp but can also involve eyebrows, eyelashes, beard hair, and body hair.
- In severe cases, patients lose all scalp hair (alopecia totalis) or all body hair (alopecia universalis).
- A scalp biopsy may be needed to confirm the diagnosis



Alopecia Areata - Treatment

- Depends on the severity of the disease.
 - Small patches often resolve spontaneously.
 - If regrowth does not occur, such patches usually respond to intralesional injections of triamcinolone acetonide
 - In more severe cases, oral JAK inhibitor (baricitinib or ritlecitinib)

Traumatic Alopecia

- Caused by a variety of physical or chemical injuries to the hair and scalp
- These injuries may be deliberate or accidental, inflicted by self or others, and acute or repetitive.
- Potential causes include trichotillomania, habit tics, pruritic dermatoses, traction, pressure and friction, heat, radiation, and chemicals.
- In most cases management consists of removal of the underlying cause.
 - In areas with permanent damage, hair transplantation may be necessary.

Traumatic Alopecia - Trichotillomania

- Compulsion to pull out one's hair.
- Characterized by an increasing sense of tension before, and a sense of relief after, the hair is pulled.
- classified as a specific disorder of impulse control
- **The diagnosis is based on the presence of irregular, broken-off hairs in patches on the scalp**
 - The hairs are irregular in length because they are broken off at different times.
 - **The scalp is normal.**
- Occasionally, biopsies are necessary to confirm the diagnosis.
- Usually, reassurance and understanding go a long way in the treatment of this condition



Traumatic Alopecia – Pruritic Dermatoses

- Pruritic dermatoses lead to hair loss from excoriation
 - acne necrotica
 - Folliculitis
 - lichen simplex chronicus
 - pediculosis capitis
 - prurigo nodularis
 - psoriasis, seborrheic dermatitis
 - neurotic excoriations
- They need to be identified and treated.

Traumatic Alopecia – traction alopecia and loose anagen syndrome

- Traction alopecia may be acute (caused by accidental or deliberate avulsion of the scalp) or may arise from a familial condition, the loose anagen syndrome.
 - Common causes - excessive brushing and combing; backcombing and pulling the hair into braids, cornrows, and ponytails; weaving; and application of rollers
- Loose anagen syndrome is usually seen in fair-haired children 2 to 5 years of age.
 - It often presents as patchy hair loss following an incident of hair tugging.
 - Prompt hair regrowth is the rule.
 - The condition becomes asymptomatic with gentle hair care.
 - Diagnosis is made on the basis of positive hair-pull tests showing many anagen hairs.

Traumatic Alopecia – Pressure and Friction

- Prolonged pressure on a localized area of the scalp in immobilized neonates or patients under anesthesia, in coma, or with a debilitating illness may result in ischemia leading to pressure alopecia.
- The hair usually regrows with time, but if the damage is severe, permanent hair loss and scarring may result



Traumatic Alopecia – heat, radiation, chemicals

- Excessive heat from hot oils and pomades, hot combs, and hot rollers is a common cause of chronic hair loss.
- Radiation dermatitis
- Chemicals
- A careful history of hair care and grooming is needed to uncover these causes

Cicatricial Alopecia

- Results from **permanent scarring of the hair follicles**
- It may be widespread or localized.
- Causes (Primary or secondary)
 - Hereditary or congenital conditions, infections, injuries, neoplasms, and dermatoses

Cicatricial Alopecia - Diagnosis

- Scarring is detected by the absence of follicular orifices and a pearly or scarred appearance of the skin.
- Associated lesions such as folliculitis, follicular plugs, scales, and telangiectasias may be found, along with broken, twisted, or easily extractable hairs.
- A scalp biopsy from an active area might lead to a specific diagnosis.

Cicatricial Alopecia – Clinical Variants

- The common variants include central centrifugal cicatricial alopecia, discoid lupus erythematosus, lichen planopilaris, folliculitis decalvans, and nonspecific cicatricial alopecia (pseudopelade).
- The end phases of these conditions are similar; they are characterized by a lack of pores and by inflammation in white, scarred areas

Table 5 Causes of Cicatricial Alopecia⁴⁰

<i>Category</i>	<i>Diagnosis</i>
Dermatoses	Cicatricial pemphigoid, dermatomyositis, folliculitis decalvans, lichen planopilaris, lupus erythematosus, neurotic excoriations, pseudopelade, scleroderma
Hereditary and congenital disorders	Aplasia cutis, epidermal nevi, epidermolysis bullosa
Infections	
Bacterial	Acne keloidalis, dissecting cellulitis, folliculitis, syphilis
Fungal	Favus (tinea capitis), kerion, mycetoma
Protozoan	Leishmaniasis
Viral	Herpes zoster, varicella
Injuries	Burns, mechanical trauma, radiodermatitis
Neoplasms	Angiosarcoma, basal cell epithelioma, lymphoma, melanoma, metastatic tumors, squamous cell carcinoma

Cicatricial Alopecia – variant Central centrifugal

- This condition usually affects middle-aged, African-American women
- Starts in the central vertex area of the scalp and gradually extends outward.
- Destructive and progressive.
- May be aggravated by physical and chemical trauma



Cicatricial Alopecia – variant Discoid Lupus

- Lesions of discoid lupus are often itchy at onset and lead to erythema, dyspigmentation, scaling, telangiectasia, follicular spines, and atrophy
- Occur anywhere on the scalp



Cicatricial Alopecia – variant Lichen planopilaris

- Starts with bare, white patches that bud out from one another like pseudopods.
- The central and vertex of the scalp are typically involved.
- Prominent follicular hyperkeratosis and erythema present around the residual terminal hairs at the edges of the lesion.
- May have concurrent mucocutaneous lesions typical of lichen planus



Cicatricial Alopecia – variant Folliculitis decalvans

- Crops of follicular pustules surrounding multiple, slowly expanding, and round or oval areas of alopecia
- May involve large areas of the skin.
- Secondary infection may be severe, with crusting and oozing.
- Eventually gradually becomes less active and looks like other forms of chronic cicatricial alopecia.



Cicatricial Alopecia – variant

Nonspecific cicatricial alopecia (pseudopelade)

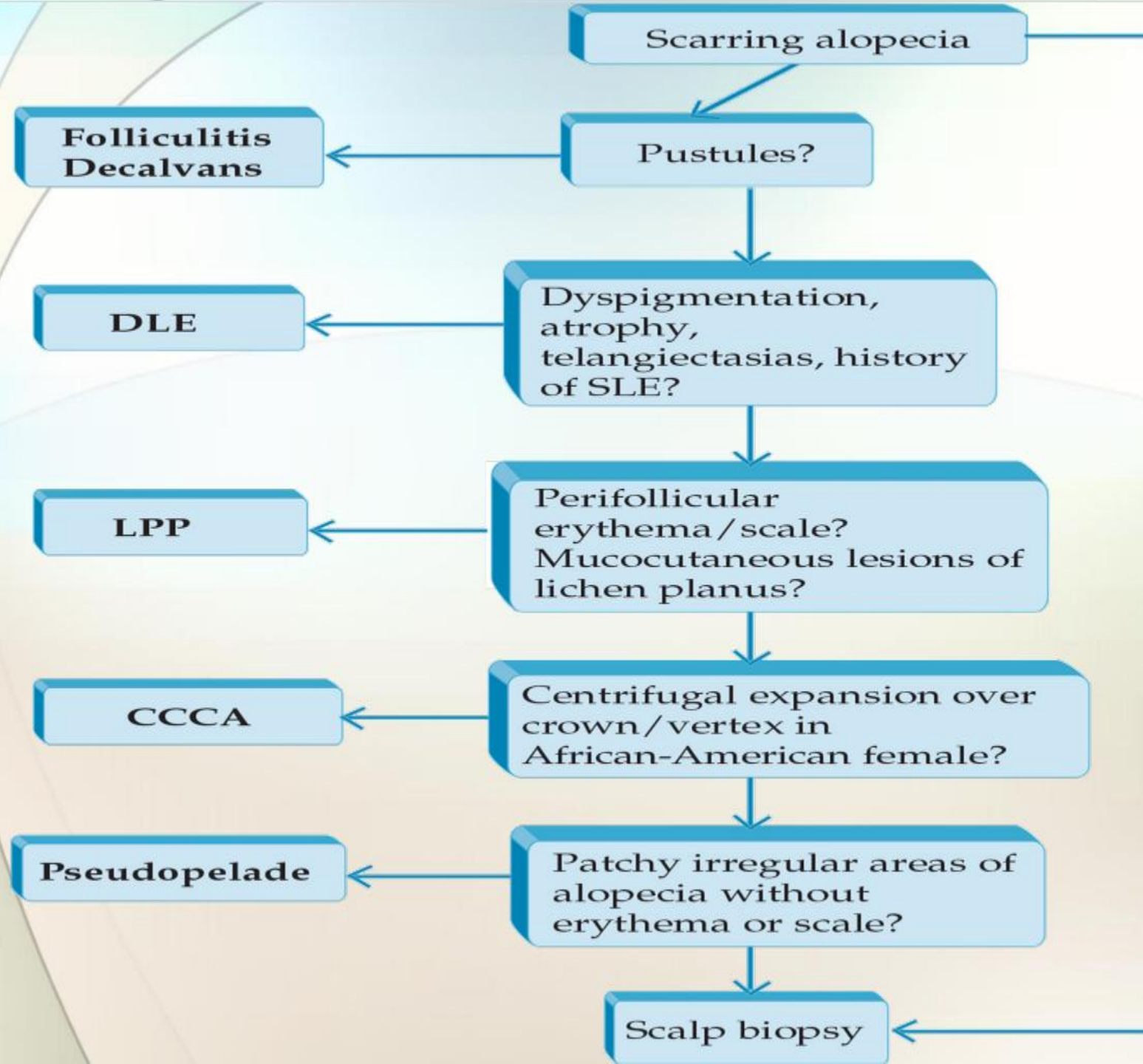
- Insidious spread of a scarring process, which is **noninflammatory**.
- Often involves the crown and occurs mainly in middle-aged women.
- **Characterized by patchy areas of alopecia with irregular extensions.**
- **The affected skin is smooth, white, and devoid of erythema, scaling, or pores**
- Course is variable and may last for a few years or several decades.
- May see it named pseudopelade de brocq (older terminology)



Dermapixel

Cicatricial Alopecia - treatment

- Depends on the level of activity of the underlying disease
- Central centrifugal cicatricial alopecia, discoid lupus erythematosus, and lichen planopilaris may respond to topical, intralesional, or systemic steroids; oral chloroquine therapy; or immunosuppressive agents such as mycophenolate mofetil or cyclosporine.
- Folliculitis decalvans may respond to treatment with long-term antibiotics such as doxycycline, minocycline, erythromycin, ciprofloxacin, cephalexin, trimethoprim-sulfamethoxazole, or rifampin
- When the conditions have burned themselves out, either scalp reduction (i.e., surgical removal of a bald area of scalp and stretching of adjacent scalp over the removed area) or hair transplants may be helpful.



Miscellaneous causes of hair loss

Table 6 Miscellaneous Causes of Hair Loss

Tinea capitis
Secondary syphilis
Scalp lice
Genetic hair shaft abnormalities
Congenital ectodermal dysplasias
Follicular mucinosis
Sarcoidosis

Miscellaneous – tinea capitis

- Largely caused by *Trichophyton tonsurans*, an endothrix that infects the inside of the hair shaft.
- Makes the shaft brittle, which causes it to snap off flush with the skin, leaving a characteristic black dot of hair.
- The clinical diagnosis depends on this finding of black dots in patchy areas of hair loss.



Miscellaneous – secondary syphilis

- Can cause a somewhat nondescript, **moth-eaten** type of diffuse alopecia
- A routine serologic test for syphilis is indicated



Miscellaneous - Lice

- Scalp lice should always be sought in cases of **hair loss accompanied by pruritus**.
- Lice are most likely to be found around and behind the ears and on the nape of the neck.
- Lymphadenopathy may also be present.
- Suitable treatment with permethrin shampoo or 0.5% malathion can be given.



Miscellaneous – hair shaft abnormalities

- Hair shaft abnormalities frequently present as broken-off hairs.
- Structural abnormalities of the hair shaft include fractures, irregularities, coiling and twisting, and extraneous matter
- They can often be diagnosed by light microscopy

Miscellaneous - congenital

- Differs from alopecia - no hair growth in the first place (vs. loss of hair)
- Congenital hypotrichosis with or without associated defects,
 - Congenital triangular alopecia, and many ectodermal dysplasias that affect the hair, teeth, nails, and sweat glands.
- One major form of congenital hypotrichosis is the Marie Unna syndrome, which affects large families that have a dominant gene
 - The hair is sparse at birth with a coarse texture, which eventually regrows in childhood with potential hair loss at puberty
- Minor forms of hypotrichosis can occur in patients with other hereditary syndromes and chromosomal abnormalities.
- In most of these conditions, a reduction of hair follicles accounts for the hair loss.

Abnormal hair growth - Hirsutism

- Idiopathic
 - Hereditary
 - Women with idiopathic hirsutism have normal menstrual cycles and no evidence of any of the conditions associated with secondary hirsutism.
- Secondary
 - hirsutism is most often associated with polycystic ovary syndrome
 - This type of hirsutism may also be caused by:
 - malfunctions of the pituitary or adrenal glands
 - use of male hormones or minoxidil
 - adrenal or ovarian tumors.
- Rarely caused by a serious underlying disorder.
 - Hair follicles usually become enlarged, and the hairs themselves become larger and darker.
 - A woman whose hirsutism is caused by an increase in male hormones has a pattern of hair growth similar to that of a man.
 - A woman whose hirsutism is not hormone-related has long, fine hairs on her face, arms, chest, and back.

Abnormal hair growth - Hirsutism

- Diagnosis is based on a family history of hirsutism, a personal history of menstrual irregularities, and masculine traits. Laboratory tests are not needed to assess the status of patients whose menstrual cycles are normal and who have mild, gradually progressing hirsutism.
- Primary hirsutism can be treated mechanically.
 - Mechanical treatment involves bleaching or physically removing unwanted hair by:
 - cutting
 - electrolysis
 - shaving
 - tweezing
 - waxing
 - using hair-removing creams (depilatories)
- Treatment is determined by the underlying cause of the condition.
- When spironolactone (Aldactone) is prescribed to suppress hair growth, 70% of patients experience improvement within six months. When women also take birth-control pills, menstrual cycles become regular and hair growth is suppressed even more.

